Fabiana S. Henríquez Portela

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Education

Duke University — Durham, NC
Bachelor of Science in Computer Science — *May 2029*
- Kinesis Foundation Scholarship Recipient (2024 - 2029)
- **Relevant Coursework:** Introduction to Computer Science, Race, Gender, Class & Computing, Calculus I & II
- **Activities:** Catalyst, Alpha Phi, Duke Responsible AI Symposium (Videographer & Editor)

Specialized Secondary School in Science, Math, and Technology (CIMATEC) — Caguas, PR
High School Diploma — *May 2024*
- 4.0 Unweighted GPA, Graduated with High Honors
- **Relevant Coursework:** Intro to Robotics, Programming Principles (C++), SolidWorks, Planetary Radar Science (Python)

Experience

Clinical Research Assistant — Jan - Oct 2025; Aug 2026 - Present
Aging Center, Duke University Hospital — *Durham, NC*
- Studying neurocognitive aging, HIV, and Parkinson's disease in Hispanic older adults under Dr. María J. Marquine at the Duke Aging Center (5 to 10 hrs/week)
- Conducting clinical research visits for Parkinson's disease and Hispanic neurocognitive studies, collecting participant data
- Administering sociodemographic and neuropsychological assessments to Hispanic older adults in North Carolina
- Managing 50+ participant records in REDCap, maintaining accurate study data for HIV neurocognition research
- Co-authored published paper on resilience and neurocognition in 116 older adults living with HIV; cleaned study dataset and conducted literature review

Production Technician — May - Aug 2026
Té K Kombucha (La Fermentería) — *Puerto Rico*
- Diagnosed over-carbonation in 200 gallons/week of kombucha production by analyzing pH, °Brix, and carbonation data, restoring consistent carbonation across weekly batches
- Tested and refined new raspberry flavor by adjusting sweetness and carbonation formulas; now sold at Walmart and Freshmart in Puerto Rico
- Built an Excel tracking system to monitor fermentation across weekly production batches

Executive Team Member & Mentor — Aug 2021 - May 2025
Latinas in STEAM — *Puerto Rico*
- Guided 3 Latina high school students into Ivy League and T10 schools with scholarship awards through coaching on essays and applications
- Organized 10+ workshops for students at CIMATEC and the University of Puerto Rico, Río Piedras as part of nonprofit leadership team
- Promoted programs and STEAM gender-gap awareness by designing monthly Facebook and Instagram posts for nearly 5K followers

Projects

Breast Tumor Diagnosis Classifier | *Python, XGBoost, scikit-learn, pandas, Google Colab*
- Built XGBoost classifier in a team of 4 to label breast tumors as benign or malignant from 30 cell-nucleus measurements
- Prepared 569-sample dataset in pandas by dropping unused columns, encoding diagnosis labels, and creating a stratified 80/20 train/test split
- Achieved 96.5% accuracy and 0.996 ROC-AUC on held-out test set, validated with 5-fold cross-validation

Personal Portfolio Website | *HTML, CSS, JavaScript, GitHub Pages*
- Built interactive portfolio styled as a Mac desktop with clickable folders, windows, and a working terminal
- Designed scrolling sections showcasing experience, research, graphic design, and photography
- Deployed with GitHub Pages using Git version control

BlackRock Asset Allocation Competition | *Portfolio Construction, Risk Analysis* — *Fall 2026*
- Built a 2065 target-date portfolio in a team of 3, shifting weight from U.S. mega-caps to international and growth sectors
- Matched BlackRock's 2065 fund Sharpe ratio (0.24) with 10.2% lower modeled volatility, backed by return forecasts for 7 ETFs

Goldman Sachs Trade Pitch Competition | *DCF, Monte Carlo Simulation, CAPM* — *Fall 2026*
- Pitched a beta-neutral long Vistra / short Constellation Energy pair trade on AI-driven power demand in a team of 3
- Built DCF and Monte Carlo valuation models to set price targets, projecting a 24.9% return on $25M over 9 months

Skills & Interests

Languages: Python, C++, HTML/CSS, JavaScript
Libraries & Tools: XGBoost, scikit-learn, pandas, Git/GitHub, Google Colab, PyCharm, REDCap, Excel
Research: Neuropsychological Assessment, Clinical Data Collection, Human Subjects Research (CITI Certified)
Creative: Video Editing (Final Cut Pro), Photography, Graphic Design
Spoken Languages: Spanish (native), English (fluent)
Interests: Soccer, Latin Music, Olivia Rodrigo, Realistic Drawing, Italian Cuisine